

involves the fact that episodic knowledge actually comprises two different things: information relating to the agent and its human collaborators, and information about real-world entities outside of the agent's world. By definition, the agent has full knowledge about its private experiences. By contrast, when it comes to public information, it can lack episodic knowledge just like it can lack ontological knowledge. However, recall what we are talking about: explaining *the agent's* knowledge. This is quite different from casting LEIAs as all-purpose question-answering systems, which they are not.

People can ask LEIAs questions about ontology or episodic knowledge either to refresh their memory or because they don't know the answer. If they are just refreshing their memory and the response sounds right, then no explanation is needed. By contrast, if the answer does not sound right, or if the information is new, then they might want to validate its veracity. To do so, they might ask questions like *How do you know?* or *What exactly did you read/hear/find?* In some cases, the agent will have recorded the source of information as metadata, as when it engages in learning by reading or is being instructed by a particular human. In cases when the source of information is not known, the agent could be instructed to search a corpus for corroborating information that includes a source; but this goes beyond the basic functionality of enabling agents to explain their current state of knowledge.

This section has described how a LEIA can explain the content of its lexicon, ontology, and episodic memory. This does not exhaust its knowledge. Another aspect of its knowledge is that which underlies its reasoning, to which we now turn.

8.4 Explaining Reasoning

According to the AGENT-FUNCTIONING-FLOW script that implements the LEIA's architecture (cf. section 3.2.4), agent action is triggered in four ways:

1. The action can be the output of a specific decision function. For example, when the agent is serving as an advisor, if it detects a user error, it issues a warning flag.
2. The action can be the ADJACENCY-PAIR of the previous event. For example, when a person asks a question, the agent answers it.

3. The action can be triggered by a daemon (a standing goal). For example, a particular virtual patient might need to know if procedures are painful before agreeing to them. If the doctor recommends a procedure for which the patient lacks information about pain, it posts the goal of tracking down that information.
4. The action can be the next step in the current plan on the agent's agenda, launched when the previous step is completed. For example, if a robotic LEIA is building a chair, then after it attaches the first leg, it undertakes to attach the second leg, and then the third, and the fourth.

In all cases, the agent knows why it chose a given action. Formally, this is recorded as metadata with the remembered instance of the event. For cases 2–4, explaining the reasoning is relatively straightforward, apart from some details of language generation. If the agent is asked *Why did you say that?* or *Why are you doing that?* it answers:

- 2a. Because [a brief description of the first event in the adjacency pair]. For example, Q: *Why did you say that?* A: *Because Joe asked me where I was going.*
- 3a. Because [a brief description of the daemon]. For example, Q: *Why did you ask that question?* A: *Because I need to know if procedures are painful before agreeing to them.*
- 4a. It's the next step in [plan name]. For example, Q: *Why are you doing that?* A: *It's the next step in building a chair.*

By contrast, explaining decision functions, the first case above, is more complicated since those decisions can involve not only agent actions but also recommendations, advice, warnings, and so on. Examples of what a person might want to know about decisions include, among many others:

- Why did you do that?
- Why did you recommend that?
- Why do you think that what I was planning to do is wrong?
- What did you base your recommendation on?
- Was your recommendation informed by machine learning?
- Would any additional information be useful in making this decision?
- How sure are you of this recommendation?
- Are you sure this is best?
- Why is [option X] better than [option Y]?

The most important point about preparing LEIAs to explain their decision-making is that **all of the information they need is either recorded as knowledge associated with the decision function or is dynamically generated and stored as metadata while the decision is being made.** This means that agents do not need to invent or reconstruct the reasoning behind their decisions if they are asked about them; they need only look up what was already prepared and then package it as a situationally appropriate utterance.

The static and dynamically generated information a LEIA relies on to explain its decision-making is recorded as values of the following metalevel properties:

- EXPL provides a short English explanation of a decision function; it typically includes variable slots.
- CONFIDENCE holds the agent's confidence in the decision, measured on the abstract scale $\{0,1\}$.
- RELEVANT-FEATURES holds the list of properties whose values, if known, contribute to the decision.
- CONTRIBUTING-FEATURE-VALUES holds the list of actual property values contributing to the decision in the given context.
- ABSENT-FEATURE-VALUES holds the list of properties for which the decision function needs values that are unknown in the given context.
- ROLE-OF-ML conveys the role of data-driven methods in the decision, apart from its role in perception processing, which is considered separately; the role of ML affects the agent's confidence in the decision.
- COMPARISON-OF-OPTIONS holds the result of comparing decision options that are above a quality threshold.
- IDIOSYNCRATIC-DECISION-TRACE allows for any other aspects of the decision function to be prepared in advance as an explanation; for example, a numerical calculation that contributes to a decision can be described in detail in plain English.

We will describe the use of these features on the example DETECT-JUMPING-TO-CONCLUSIONS-DIAGNOSIS, which is used by agents that are serving as medical tutors or advisors. This algorithm detects whether a diagnosis posited by a system user is clinically appropriate. Here we focus on how this algorithm's content supports explanation; how the algorithm is used in a medical application system is described in section 8.5.